

TraceLens AI — Hackathon Submission Kit

Complete Official Proposal, Slide Deck, Video Script & Technical Rubric Alignment

1. Project Title & Descriptions

Project Title:	TraceLens AI: Autonomous Microservice Incident Triage & Root-Cause Localization Engine
Short Description: (Tagline < 130 chars)	Graph-powered AIOps engine that pinpoints microservice bottlenecks via DAG Critical Path analysis and auto-generates SRE remediation scripts.
Long Description:	Modern cloud-native architectures generate billions of asynchronous trace spans and unstructured logs across hundreds of microservices. When a production incident occurs, Site Reliability Engineers (SREs) spend up to 70% of their Mean Time to Resolution (MTTR) manually correlating distributed telemetry across siloed dashboards. This manual triage costs enterprise organizations an average of \$9,000 per minute of downtime. TraceLens AI bridges this critical gap through a novel, hybrid intelligence pipeline: • Algorithmic Graph Ingestion: Ingests OpenTelemetry and Jaeger distributed traces and models them as asynchronous Directed Acyclic Graphs (DAGs). • Deterministic Critical-Path & Slack Analysis: Computes early/late task schedules to mathematically isolate latency bottlenecks and cascading failure propagation paths. • Edge-Conditioned Graph Attention Networks (GATConv): Analyzes node-level durations and inter-service network transit delays to score anomaly severity and root-cause probabilities. • Agentic LLM Triage & Remediation: Synthesizes graph telemetry and contextual error logs to generate executive incident summaries, root-cause diagnostics, and copy-paste-ready remediation scripts (kubectl, SQL, and Bash commands). TraceLens reduces triage time from 45 minutes to under 5 seconds, delivering predictable cloud reliability and enterprise-grade resilience.
Categories & Tags:	Category: AIOps / Developer Tools / Cloud Infrastructure & Observability / Enterprise AI Tags: Graph Neural Networks, OpenTelemetry, Large Language Models, Distributed Tracing, AIOps, Streamlit, Root Cause Analysis, FastAPI, Site Reliability Engineering (SRE)

2. Slide Deck Outline (16:9 Presentation Format)

Slide #	Title	Content & Visual Blueprint
Slide 1	TraceLens AI	Title, Tagline, Team Members, Live Demo QR Code / Platform URL.
Slide 2	The Crisis: Downtime & SRE Fatigue	• \$9,000/min enterprise downtime cost. • 70% MTTR wasted on manual triage. • Existing APMs (Datadog/Dynatrace) alert on symptoms, not root cause.
Slide 3	The Solution: Hybrid Graph AI	Architecture diagram: Raw OTel Traces → DAG Slack Analysis → Edge-GAT Conv → Groq LLM Remediation Agent.
Slide 4	Technical Innovation & Uniqueness	Deterministic DAG Critical Path pre-filtering + Edge-Conditioned GAT representation vs simple prompt wrappers.
Slide 5	Product Demo Highlights	Screenshots of Streamlit UI: Chaos injection button, waterfall trace graph, and auto-generated kubectl fix script.
Slide 6	Market Analysis & Business Model	• TAM: \$61.3B Observability Market by 2028 (11.7% CAGR). • Tiered B2B SaaS: Developer (Free), Team (\$49/seat), Enterprise (\$0.05/1K spans).
Slide 7	Competitive Matrix	Comparison table vs Datadog, Dynatrace, and generic LLMs across 4 core dimensions.

Slide 8	Roadmap & Future Vision	• Q3 2025: Kubernetes Operator for automated self-healing. • Q4 2025: eBPF zero-code kernel instrumentation.
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3. Video Presentation Script (Target: 3:30 – 4:15 Min)

Structured specifically to satisfy the 3 to 5 minute mandatory video criteria.

0:00 – 0:45 The Hook & Problem	"Every minute of enterprise cloud downtime costs an estimated \$9,000. In distributed microservice environments with hundreds of interconnected containers, when an incident hits, alarms fire everywhere at once. SREs and on-call engineers spend 45 minutes digging through gigabytes of logs and thousands of trace spans just to find out which microservice broke first. This is called the Alert Storm problem—where engineers see the symptoms, but not the root cause."
0:45 – 1:30 Innovation & Tech	"Meet TraceLens AI—an autonomous incident triage and root-cause localization platform. Unlike generic AI wrappers that dump unstructured text into an LLM, TraceLens uses a novel hybrid architecture. First, it converts raw OpenTelemetry traces into asynchronous Directed Acyclic Graphs. Second, our algorithmic engine runs critical-path and float-time analysis to isolate bottlenecks. Third, our Edge-Conditioned Graph Attention Network pinpoints the exact culprit span. Finally, an LLM agent generates an executive incident report along with ready-to-run remediation code."
1:30 – 3:00 Live UI Demo Walkthrough	"Let's see it live. Here on our Streamlit dashboard, we can ingest OpenTelemetry or Jaeger traces. I'll click our Trigger Simulated Outage button to simulate a database lock that cascades into a payment service timeout. Within 200 milliseconds, TraceLens isolates the database span on the critical path with a 98% anomaly confidence score. On the right, the AI generates the complete SRE report—identifying connection pool exhaustion—and provides the exact kubectl restart command and PostgreSQL pool configuration patch to resolve it instantly."
3:00 – 3:45 Business, Market & Roadmap	"The Observability market is projected to reach over \$61 Billion by 2028. TraceLens operates on a B2B SaaS model with per-node pricing and enterprise integrations into Slack, PagerDuty, and Datadog. Our next milestone is closed-loop autonomous remediation: allowing verified Kubernetes operators to execute safe rollback patches automatically. TraceLens transforms on-call engineering from high-stress firefighting to automated 5-second resolutions. Thank you!"

4. Business Value & Market Analysis

Total Addressable Market (TAM):	• TAM: Global Cloud Observability & AIOps: \$61.3 Billion by 2028 (CAGR 11.7%) • SAM: Cloud-Native Kubernetes & Microservices Observability: \$14.2 Billion • SOM: Mid-to-Enterprise DevOps teams running distributed architectures: \$1.8 Billion
Revenue Model:	• Developer Tier (Free / OSS): Up to 10k spans/month with local CLI & dashboard. • Team Tier (\$49/seat/mo): Real-time Groq LLM triage, PagerDuty/Slack incident bot. • Enterprise Tier (\$0.05 / 1k spans): Custom on-prem GNN inference, SSO & auto-remediation.

Competitive Advantage Matrix:

Feature	Datadog / Dynatrace	Pure LLM Wrappers	TraceLens AI (Ours)
Root-Cause Localization	Rule/Threshold based (High noise)	Flawed (No graph context)	Mathematical DAG Slack + Edge GNN
Telemetry Ingestion	Proprietary Agents (Expensive)	Text logs only	Native OpenTelemetry & Jaeger
Actionable Fix Scripts	No (Manual Dashboard)	Generic hallucinations	Context-aware kubectl / SQL scripts
Mean Time to Triage	15 - 45 Minutes	2 - 5 Minutes	< 5 Seconds

■ 5. Scoring Criteria Alignment Checklist (Target: 5 / 5)

Criteria	Target Score	Demonstrated Evidence in Project
1. Presentation (PDF & Video)	5 / 5 (Excellent)	• Flawlessly communicates problem, solution, and value prop in < 5 min video. • Comprehensive 16:9 PDF deck with market sizing (TAM/SAM/SOM), revenue model, and roadmap.
2. Business Value	5 / 5 (Exceptional)	• Directly addresses \$9k/minute cloud downtime problem. • Reduces MTTR by 85% with clear enterprise ROI. • Sustainable multi-tier B2B SaaS business model.
3. Tech Application	5 / 5 (Excellent)	• Production-ready hybrid pipeline: DAG Critical Path + PyTorch Edge-Conditioned GAT + Groq LLM. • Interactive Streamlit web app and clean GitHub repository with automated tests.
4. Originality	5 / 5 (Transformative)	• Unprecedented combination of deterministic graph theory with deep neural embeddings to eliminate LLM hallucinations and scale across 1,000+ spans.

■ 6. Next Steps to Deploy Your Demo

1. Deploy to Streamlit Cloud:

- Push project code to a public GitHub repository.
- Go to share.streamlit.io, connect your repo, and add GROQ_API_KEY under **App Settings** → **Secrets**.
- Copy your live public URL for the submission form.

2. Generate Slides & Video:

- Run this script to generate your official submission PDF.
- Record your 3-4 minute walkthrough with Loom or OBS using the script above and export to MP4.